

PRISM: A Pedagogical Resource Inspection System for Multimodal Educational Content Quality Evaluation

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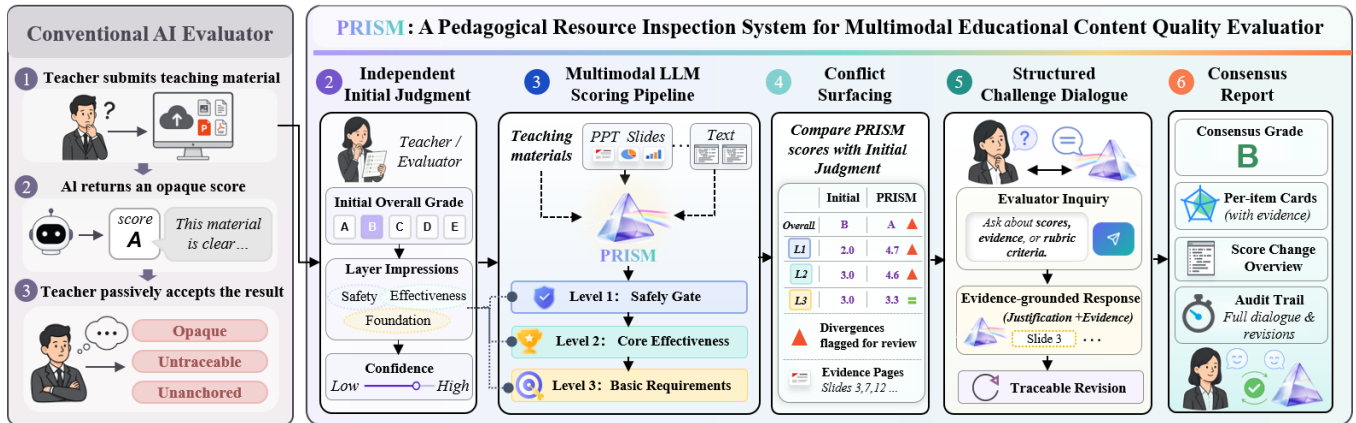


Figure 1: System overview. A conventional AI evaluator (left) returns an opaque score the teacher passively accepts; PRISM (right) scaffolds a five-stage workflow: independent initial judgment, multimodal LLM scoring against a three-layer non-compensatory rubric, conflict surfacing, structured challenge dialogue, and an audited consensus report.

Abstract

We present PRISM (Pedagogical Resource Inspection System for Multimodal content), an interactive system for evaluating AI-generated and teacher-created multimodal educational resources. PRISM grounds evaluation in a validated 24-indicator three-layer non-compensatory rubric: a safety gate with veto authority, a core effectiveness layer yielding a five-tier grade, and a diagnostic layer. A multimodal

LLM jointly processes slide images and text with page-level evidence. Beyond "show-and-accept" scoring, PRISM supports traceable human-AI challenge and revision. Evaluators record an independent initial judgment before PRISM scores are revealed and may contest any score through traceable dialogue. Project page: <https://chenzhangze-web.github.io/prism-page/>.

CCS Concepts

• Computing methodologies → Artificial intelligence; • Applied computing → Education.

Keywords

multimodal LLM; educational resource evaluation; human-in-the-loop; non-compensatory rubric

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1 Introduction

Generative AI has compressed lesson-plan creation from hours to minutes, but the quality of AI-generated materials varies widely in accuracy, pedagogical soundness, and ethical compliance [9]. Applying a general-purpose LLM as evaluator produces holistic judgments that conflate safety-critical failures (fabricated citations, undisclosed AI authorship) with mild pedagogical weaknesses; rubric-guided LLM scoring improves alignment with human judges but does not enforce safety asymmetry [3, 4]. Neither approach addresses the multimodal nature of modern teaching materials, where image-text consistency and media quality require joint visual-linguistic reasoning that recent multimodal LLM research has begun to address for documents [5, 11] and unified image-text understanding [6].

PRISM contributes (1) a theory-grounded 24-indicator rubric that enforces a non-compensatory veto so safety-critical failures cannot be offset by pedagogical strengths; (2) a multimodal LLM scoring engine that reasons jointly over slide images and text with page-level evidence; and (3) a five-dimensional challenge mechanism, building on proactive conversational AI [2], in which an *independent initial judgment* seeds a structured-challenge dialogue, operationalising appropriate-reliance principles for educational-resource evaluation (Figure 1).

2 Evaluation Framework

PRISM organises evaluation along a 24-indicator three-layer non-compensatory rubric (Figure 1), grounded in Merrill’s First Principles [8], Mayer’s CTML [7], the ICAP framework [1], and UNESCO AI-in-education guidelines [9], and validated through a Modified Delphi study with 15 domain experts. **L1 Safety Gate** (6 indicators covering factual accuracy, ethics, privacy, link safety, citation verifiability, and AI-authorship disclosure) screens for content correctness and ethical compliance: any L1 score ≤ 2 triggers an immediate “prohibited from use” verdict regardless of L2/L3 performance, operationalising the fatal-flaw principle [10]. **L2 Core Effectiveness** (11 indicators on teaching aims, content coverage, learning activities, cognitive design, and assessment) is the grade-determining layer, with the indicator mean mapped to grades A/B/C/D/E (thresholds 4.5 / 3.5 / 2.5 / 1.5). **L3 Basic Requirements** (7 indicators on visual design, readability, media quality, metacognitive support, and technical usability) contributes diagnostic detail without affecting the grade. This non-compensatory layering is the key architectural decision distinguishing PRISM from holistic LLM scoring, which has no equivalent veto.

3 System Design

Architecture and Workflow. PRISM is implemented as a Next.js front end with a Python/FastAPI back end. The five-step workflow proceeds as: (i) resource upload with metadata, used to conditionally activate indicators; (ii) *independent initial judgment* (overall A to E grade, three-layer impressions, and a confidence rating) recorded *before* any AI output is shown; (iii) three sequential LLM rounds, one per layer, with real-time progress; (iv) a comparison

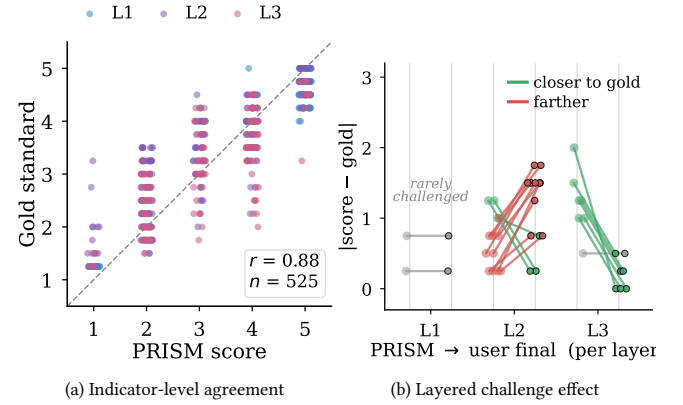


Figure 2: Pilot evaluation by rubric layer. (a) PRISM vs. 4 experts gold-standard scores. (b) Challenge-induced shifts from PRISM to user-final scores.

view that surfaces conflicts and supports the challenge phase (Section 3); (v) a final report combining overall-grade cards, a three-column qualitative summary (strengths, weaknesses, suggestions), and per-indicator detail panels with embedded challenge records.

Multimodal LLM Scoring. Each evaluation round receives *both* slide images and extracted text, since indicators on multimodal consistency, visual quality, and readability cannot be judged from text alone. Three sequential LLM calls (one per layer) each receive layer-specific rubric prompts with 1 to 5 behavioral anchors per indicator and return structured JSON (code, score, comment, evidence_pages). Separating layers keeps each layer independently auditable.

Five-Dimensional Challenge Mechanism. After AI scoring, PRISM enters a challenge phase organised along five dimensions (Figure 1): (1) **Conflict Surfacing** highlights divergences ≥ 1 grade level or ≥ 1.5 points; (2) **Challenge Modality** offers structured and free-form contestation; (3) **Response Strategy** requires each revision to be grounded in concrete evidence, such as a specific slide page and rubric anchor; (4) **Authority Locus** preserves both AI and user-final scores for audit; (5) **Learning Loop** retains challenge history, surfacing recurring revision targets.

4 Preliminary Evaluation

We conducted a pilot study with $N=12$ educators (3 in-service teachers, 9 graduate students), each evaluating two stage-matched resources from a curated 12-resource subset covering 5 quality tiers (A to E) and 8 subject domains. The gold standard was set by two parallel 4-expert panels with combined ICC[2,k] = 0.919. PRISM matched the gold standard exactly on 62.5% of sessions and within ± 1 grade tier on all of them; per-indicator MAE was 0.643 (Pearson $r=0.88$, $p < .001$). Validity varies systematically across layers (L1: 0.380; L2: 0.676; L3: 0.869). The 22 challenge-induced revisions diverge sharply across layers (Figure 2): L3 revisions are uniformly corrective (7/7 closer to gold), L2 revisions skew toward user over-strictness (9/12 farther), and L1 is essentially unchallenged. SUS averaged 82.92 (Good on Bangor’s scale) and TAM subscales 5.96 to 6.08 on a 7-point scale.

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References

- [1] Michelene T. H. Chi and Ruth Wylie. 2014. The ICAP Framework: Linking Cognitive Engagement to Active Learning Outcomes. *Educational Psychologist* 49, 4 (2014), 219–243. doi:10.1080/00461520.2014.965823
- [2] Yang Deng, Lizi Liao, Wenqiang Lei, Grace Hui Yang, Wai Lam, and Tat-Seng Chua. 2025. Proactive Conversational AI: A Comprehensive Survey of Advancements and Opportunities. *ACM Transactions on Information Systems* 43, 3 (2025), 67:1–67:45. doi:10.1145/3715097
- [3] Helia Hashemi, Jason Eisner, Corby Rosset, Benjamin Van Durme, and Chris Kedzie. 2024. LLM-Rubric: A Multidimensional, Calibrated Approach to Automated Evaluation of Natural Language Texts. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Bangkok, Thailand, 13806–13834. doi:10.18653/v1/2024.acl-long.745
- [4] Joy He-Yueya, Noah D. Goodman, and Emma Brunskill. 2024. Evaluating and Optimizing Educational Content with Large Language Model Judgments. In *Proceedings of the 17th International Conference on Educational Data Mining (EDM 2024)*. International Educational Data Mining Society, 68–82. arXiv:2403.02795
- [5] Haoxuan Li, Wei Song, Aofan Liu, and Peiwu Qin. 2025. AdaDocVQA: Adaptive Framework for Long Document Visual Question Answering in Low-Resource Settings. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*. ACM, Dublin, Ireland, 13777–13783. doi:10.1145/3746027.3761996
- [6] Yiyang Ma, Xingchao Liu, Xiaokang Chen, Wen Liu, Chengyue Wu, Zhiyu Wu, Zizheng Pan, Zhenda Xie, Haowei Zhang, Xingkai Yu, Liang Zhao, Yisong Wang, Jiaying Liu, and Chong Ruan. 2025. JanusFlow: Harmonizing Autoregression and Rectified Flow for Unified Multimodal Understanding and Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 7739–7751.
- [7] Richard E. Mayer. 2021. *Multimedia Learning* (3rd ed.). Cambridge University Press, Cambridge, UK. doi:10.1017/9781316941355
- [8] M. David Merrill. 2002. First Principles of Instruction. *Educational Technology Research and Development* 50, 3 (2002), 43–59. doi:10.1007/BF02505024
- [9] Fengchun Miao and Wayne Holmes. 2023. *Guidance for Generative AI in Education and Research*. UNESCO, Paris. https://unesdoc.unesco.org/ark:/48223/pf0000386693
- [10] Michael Scriven. 1991. *Evaluation Thesaurus* (4th ed.). Sage Publications, Newbury Park, CA.
- [11] Dongsheng Wang, Natraj Raman, Mathieu Sibue, Zhiqiang Ma, Petr Babkin, Simerjot Kaur, Yulong Pei, Armineh Nourbakhsh, and Xiaomo Liu. 2024. DocLLM: A Layout-Aware Generative Language Model for Multimodal Document Understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Bangkok, Thailand, 8529–8548. https://aclanthology.org/2024.acl-long.463/